

SQL Assessment Tasks for Data Analytics Working Student

Dataset Overview

You will work with the following tables:

1. orders

Contains individual order records, including the customer, date, product, and country.

Columns: order_id, customer_id, order_date, product_number, country_iso

2. marketing_sources

Information about marketing channels associated with orders.

Columns: country_iso, order_id, ordertime, utm_campaign, reporting_channel

3. product_locale

Contains localized product names.

Columns: sku, product_name, locale

4. product_universal

Contains universal product information such as categories.

Columns: sku, main_category

All queries must be written using **BigQuery SQL syntax**.

Task 1: Customer Order Activity and New Customer Share Over Time

Objective: Understand early customer behavior and analyze the share of first-time buyers over time.

Instructions:

- For each customer, determine their first order date.
- Calculate how many orders they placed within 10, 15, and 20 days after their first order.
- Use ROW_NUMBER() partitioned by customer_id to identify each customer's first order.
- For each order_date, compute:
 - Total number of orders.
 - Number of orders that were the customer's first ever.
 - Percentage of total orders made by new customers.
- **Bonus:** Add a breakdown by reporting_channel from the marketing_sources table.

Task 2: Product Popularity by Country

Objective: Identify popular and unpopular products and categories by geography.

Instructions:

- For each country_iso, calculate:
 - Total number of orders and skus by main_category.
- Use RANK() or DENSE_RANK() to:
 - Identify the top 5 most ordered and bottom 5 least ordered products per country.

Task 3: Data Visualization of Key Insights

Objective: Communicate your findings visually using charts and graphs.

Instructions:

- Using the results from Tasks 1 and 2, create at least **two visualizations** of your choice. Examples include:
 - A line chart showing the daily percentage of new customers over time.
 - A bar chart comparing top product categories by order volume across countries.
 - A heatmap of new customer share by reporting_channel and time period.
 - A ranked bar chart of the top 5 and bottom 5 products per country.
- Your visualizations should include proper titles, labeled axes, and a legend where applicable.
- Briefly explain (2–3 sentences per chart) what insight the visualization communicates and why you chose that chart type.
- You may use any visualization tool or library of your choice (e.g. Python matplotlib/seaborn/plotly, Tableau, Looker Studio, Excel).

Tips

- Be sure to handle potential duplicates, nulls or nan values.
- Use SAFE_CAST or PARSE_DATE as needed for date operations in BigQuery.
- You may need to join multiple tables for full context in each task.